

On an endomorphism conjecture of Fano manifolds of Picard number 1

(Joint works with Feng Shao)

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1. Background, Related Questions & Previous results

- Throughout this talk, we work over $\mathbb{C}$.
- Let X be a projective variety. The **Picard number** of X is defined as $\rho(X) := \dim_{\mathbb{R}} \mathrm{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$ where $\mathrm{NS}(X) = \mathrm{Pic}(X)/\mathrm{Pic}^{\circ}(X)$ is the **Néron-Severi group** of X .
- A smooth projective variety X is said to be **Fano**, if the anti-canonical divisor $-K_X$ is ample.
- The **(Fano) index** of a smooth Fano variety X is defined as the maximal integer $i(X)$ such that $-K_X \sim i(X)H$ for some integral ample divisor H (called the **fundamental divisor** of X). It is known that $i(X) \leq \dim(X) + 1$. If $i(X) = \dim(X) + 1$, then $X \cong \mathbb{P}^n$; if $i(X) = \dim(X)$, then $X \cong \mathbb{Q}^n$.
- A Fano manifold X is said to be **del Pezzo** (resp. **Mukai**) if $i(X) = \dim(X) - 1$ (resp. $i(X) = \dim(X) - 2$). There is a full classification of Fano manifolds of Fano index $\geq \dim(X) - 2$.

1. Background, Related Questions & Previous results

Our work is inspired from the following folklore conjecture.

Endomorphism- $\mathbb{P}^n$ -Conjecture

If a smooth Fano variety of Picard number 1 admits a non-isomorphic surjective endomorphism, then it has to be a projective space.

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Endomorphism- $\mathbb{P}^n$ -Conjecture

If a smooth Fano variety of Picard number 1 admits a non-isomorphic surjective endomorphism, then it has to be a projective space.

The above conjecture has been verified in the following cases.

- ① (Paranjape-Srinivas, 89), (Hwang-Nakayama, 11) (Almost) homogeneous spaces
- ② (Amerik-Rovinsky-Van de Ven, 99), (Hwang-Mok, 03) Fano 3-folds
- ③ (Paranjape-Srinivas, 89), (Beauville, 01) smooth hypersurfaces of $\mathbb{P}^n$
- ④ (Hwang-Mok, 03) X contains a rational curve with trivial normal bundle
- ⑤ (Shao-Z., 22, 23) & (Kawakami-Totaro, 23) Fano manifolds with Fano index $\geq \dim(X) - 2$

1. Background, Related Questions & Previous results

The Endomorphism- $\mathbb{P}^n$ -Conjecture is indeed a fundamental building block of the following more general question.

Bounded-Degree-Conjecture (Van de Ven's; Amerik, 97)

Let $f : Y \rightarrow X$ be a finite morphism between smooth projective varieties of $\dim n \geq 2$ and of $b_2(Y) = b_2(X) = 1$. Suppose $X \not\cong \mathbb{P}^n$. Then $\deg(f)$ is bounded in terms of the discrete invariants of Y and X .

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The conjecture is known in the following cases:

- X is of general type (Kobayashi-Ochiai, 75)
- X is strict Calabi-Yau (Amerik-Rovinsky-Van de Ven, 99)
- (Amerik, 97; Amerik-Rovinsky-Van de Ven, 99) $\dim(X) \leq 3$
- (Beauville, 01), (Amerik, 07), (Shao-Z., 22) X is a smooth hypersurf. of $\dim(X) \geq 3$
- (Shao-Z., 22), (Kawakami-Totaro, 23) Fano fourfolds with index ≥ 2
- (Shao-Z., 22), (Kawakami-Totaro, 23) Del Pezzo manifolds

1. Background, Related Questions & Previous results

Definition (Bott-vanishing)

A smooth projective variety X is said to satisfy the *Bott-vanishing* if $H^j(X, \Omega_X^i \otimes L) = 0$ for every $j > 0$, $i \geq 0$, and L ample line bundle.

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- (Kawakami-Totaro, 23) proved that, the Bounded-Degree-Conjecture holds for those X violating Bott-vanishing.
- Endomorphism $\Rightarrow (\times)$ B-D property $\Rightarrow$ Bott-vanishing
- Bott proved Bott-vanishing for projective spaces.
- Danilov and Steenbrink proved that every smooth projective toric varieties satisfies Bott-vanishing.

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- Bott proved Bott-vanishing for projective spaces.
- Danilov and Steenbrink proved that every smooth projective toric varieties satisfies Bott-vanishing.
- Totaro found a non-toric example satisfying Bott-vanishing: quintic del Pezzo surface.
- A result of Nakayama showed that such a non-toric surface admits no non-isomorphic surjective endomorphism.
- The converse direction of the red arrows is in general not true.

1. Background, Related Questions & Previous results

As a higher Picard number analogue, Fakhruddin asked the following.

Dynamic-Toric-Conjecture (Fakhruddin, 03), (Meng-Z., 23)

Let X be a rationally connected smooth projective variety. Suppose X admits an int-amplified endomorphism. Is X a toric variety?

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Let X be a rationally connected smooth projective variety. Suppose X admits an int-amplified endomorphism. Is X a toric variety?

- A surj. endo. $f : X \rightarrow X$ of a proj. variety X is **int-amplified** if one of the following equivalent conditions holds (Meng-Zhang, 18), (Meng, 20), (Meng-Zhang, 23).
 - ▶ $f^*H - H$ is ample for some ample Cartier divisor H on X ;
 - ▶ All of the eigenvalues of $f^*|_{N^1(X)}$ have modulus > 1 ;
 - ▶ The last dynamical degree of f is strictly larger than the other dynamical degrees.

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 - ▶ All of the eigenvalues of $f^*|_{N^1(X)}$ have modulus > 1 ;
 - ▶ The last dynamical degree of f is strictly larger than the other dynamical degrees.
- A smooth projective toric variety of Picard number one is $\mathbb{P}^n$. Any non-isomorphic surjective endomorphism of a Fano manifold of Picard number 1 has to be int-amplified.
- Toric varieties having many endomorphisms, and they violate B-D.

1. Background, Related Questions & Previous results

As a higher Picard number analogue, Fakhruddin asked the following.

Question (Fakhruddin, 03); cf. (Meng-Z., 23)

Let X be a rationally connected smooth projective variety. Suppose X admits a polarized or an int-amplified endomorphism f . Is X toric?

This question has a positive answer in the following cases:

- (Nakayama, 02) $\dim(X) = 2$.
- (Hwang-Nakayama, 11), (Meng-Zhang, 19), (Meng-Z., 23) The ramification is totally invariant. Recall that a closed subset $Z \subseteq X$ is f^{-1} -invariant or f -totally invariant for a surjective endomorphism $f : X \rightarrow X$ if $f^{-1}(Z) = Z$.
- (Meng-Zhang-Z., 22) X is a Fano threefold.
- (Jia-Z., 23) X is a Fano fourfold admitting a conic bundle structure.
- (Shao-Z., 23) X is a Fano manifold of index $\geq \dim(X) - 2$.

2. Main result & Strategy towards the proof

Target- $\mathbb{P}^n$ -Question (Shao-Z., 23)

Let X be a Fano manifold of Picard number 1 whose tangent bundle T_X is big, i.e., $\mathcal{O}_{\mathbb{P}(T_X)}(1)$ is big. Then for any non-isomorphic surjective morphism $X \rightarrow Y$ onto a smooth projective variety Y , we have $Y \cong \mathbb{P}^n$.

- (Hwang-Mok, 99); cf. (Lazarsfeld, 84) X rational homogeneous (✓)
- (Höring-Liu, 23), (HM, 99); cf. (Shao-Z., 23) Fano threefolds (✓)

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Without the bigness of the tangent bundle, the above question (✗)

- Given integers $d_1 \geq \cdots \geq d_n \geq 2$, there exists a smooth complete intersection X of multi-degree $(d_1, \dots, d_n)$ which admits a degree d_n finite cover onto a smooth complete intersection of multi-degree $(d_1, \dots, d_{n-1})$.
- The tangent bundle of X is verified non-big (Höring-Liu-Shao, 22), (Beauville-Etesse-Höring-Liu-Voisin, 23), (Liu, 23), (Shao-Z., 23).

2. Main result & Strategy towards the proof

Theorem (Shao-Z., 23)

Let X and Y be Fano manifolds of Picard number 1. Assume that the VMRT $\mathcal{C}_x \subseteq \mathbb{P}(\Omega_{X,x})$ (resp. $\mathcal{C}_y \subseteq \mathbb{P}(\Omega_{Y,y})$) at a general point $x \in X$ (resp. $y \in Y$) is not dual defective. If the tangent bundle T_X is big (i.e., $\mathcal{O}_{\mathbb{P}(T_X)}(1)$ is big), then any surjective morphism $X \rightarrow Y$ is an isomorphism unless Y is a projective space; in particular, X admits no non-isomorphic surjective endomorphism unless it is a projective space.

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- The closure of the set of all tangent hyperplanes of a projective variety $Z \subseteq \mathbb{P}^N = \mathbb{P}(V)$ is called the **dual variety** $\check{Z} \subseteq \mathbb{P}^N = \mathbb{P}(V^*)$, where V^* is the dual vector space of V . We say that Z is **dual defective**, if $N - 1 - \dim(\check{Z}) > 0$ holds.

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- The non-dual defective assumption is not very restrictive: e.g., any Fano threefold of Picard number 1 has a dominating family of minimal rational curves whose VMRT along a general point is not dual defective.

2. Main result & Strategy towards the proof

We briefly review the theory of minimal rational curves.

- Let X be a uniruled projective manifold and $\text{RatCurves}^n(X)$ the normalized space of rational curves on X .
- An irreducible component $\mathcal{K}$ of $\text{RatCurves}^n(X)$ is called a **family of rational curves**.

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- Let X be a uniruled projective manifold and $\text{RatCurves}^n(X)$ the normalized space of rational curves on X .
- An irreducible component $\mathcal{K}$ of $\text{RatCurves}^n(X)$ is called a **family of rational curves**.
- A **dominating family of minimal rational curves** $\mathcal{K}$ is an irreducible component of $\text{RatCurves}^n(X)$ such that the locus $\text{Locus}(\mathcal{K})$ of X swept out by curves from $\mathcal{K}$ is dense in X and for a general point $x \in \text{Locus}(\mathcal{K})$ the closed subset $\mathcal{K}_x \subseteq \mathcal{K}$ parametrizing curves through x is proper.

$$\begin{array}{ccc} \mathcal{U} & \xrightarrow{e} & X \\ q \downarrow & & \\ & \mathcal{K} & \end{array}$$

- It is known that any uniruled projective manifold carries a dominating family of minimal rational curves.

2. Main result & Strategy towards the proof

- A general element $[\ell] \in \mathcal{K}$ is called a **standard rational curve**, i.e., there exists a non-negative integer p such that

$$f^* T_X \cong \mathcal{O}_{\mathbb{P}^1}(2) \oplus \mathcal{O}_{\mathbb{P}^1}(1)^{\oplus p} \oplus \mathcal{O}_{\mathbb{P}^1}^{\oplus (\dim(X)-p-1)}$$

where $f: \mathbb{P}^1 \rightarrow \ell$ is the normalization.

- Let $x \in X$ be a general point, and $\mathcal{K}_x^n$ the normalization of $\mathcal{K}_x$, which is a finite union of smooth projective varieties of dimension p .

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where $f: \mathbb{P}^1 \rightarrow \ell$ is the normalization.

- Let $x \in X$ be a general point, and $\mathcal{K}_x^n$ the normalization of $\mathcal{K}_x$, which is a finite union of smooth projective varieties of dimension p .
- There is a tangent map $\tau_x: \mathcal{K}_x^n \dashrightarrow \mathbb{P}(\Omega_{X,x})$ sending a curve that is smooth at x to its tangent direction at x .
- Let $\mathcal{C}_x$ be the image of τ_x , which is called the **variety of minimal rational tangents** (VMRT) at x associated to $\mathcal{K}$. By (Kebekus, 02) and (Hwang-Mok, 04), the map τ_x is the normalization morphism of $\mathcal{C}_x$ and hence the dimension of $\mathcal{C}_x$ is equal to the dimension of $\mathcal{K}_x$ which is exactly p , see (Kollár, 96).

2. Main result & Strategy towards the proof

- For a general $[\ell] \in \mathcal{K}$, a **minimal section** $\bar{\ell}$ of $\mathbb{P}(T_X)$ over the curve ℓ is a section which is given by a surjection $f^*T_X \rightarrow \mathcal{O}_{\mathbb{P}^1}$. If X is not $\mathbb{P}^n$, then a minimal section $\bar{\ell}$ always exists, and $\mathcal{O}_{\mathbb{P}(T_X)}(1) \cdot \bar{\ell} = 0$.

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- The **total dual VMRT** of $\mathcal{K}$ is defined as

$$\check{\mathcal{C}} := \overline{\bigcup_{\text{general } [\ell] \in \mathcal{K}} \bar{\ell}}^{\text{Zar}} \subseteq \mathbb{P}(T_X)$$

where the union is taken over all minimal sections over all standard rational curves in $\mathcal{K}$.

- Let $\check{\mathcal{C}}_x$ be the fibre of $\check{\mathcal{C}} \rightarrow X$ at $x \in X$. From (Muñoz, Occhetta, Solá Conde, Watanabe, Wiśniewski, 15) and (Occhetta, Solá Conde and Watanabe, 16), $\check{\mathcal{C}}_x$ is the dual variety of $\mathcal{C}_x$ and hence the total dual VMRT $\check{\mathcal{C}} \subseteq \mathbb{P}(T_X)$ is a prime divisor if and only if for a general point $x \in X$, $\mathcal{C}_x \subseteq \mathbb{P}(\Omega_{X,x})$ is not dual defective.

2. Main result & strategy towards the proofs

The following theorem is crucial to the proof of our main theorem and has its independent interests.

Theorem (Shao-Z., 23)

Let $f: X \rightarrow Y$ be a finite morphism between Fano manifolds of Picard number 1. Let $\mathcal{K}$ and $\mathcal{G}$ be the dominating families of minimal rational curves on X and Y whose VMRTs along a general point are not dual defective. Suppose that Y is not isomorphic to a projective space and the induced rational map $\mathbb{P}(T_X) \dashrightarrow \mathbb{P}(T_Y)$ sends the total dual VMRT of $\mathcal{K}$ to the total dual VMRT of $\mathcal{G}$. Then f is an isomorphism.

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- Assume that Y is not isomorphic to a projective space. Then X is not isomorphic to a projective space, either (Lazarsfeld, 84).
- Making the above theorem as a black box, **we only need to verify that when T_X is big, the total dual VMRT is sent to total dual VMRT via the induced dominant rational map $\mathbb{P}(T_X) \dashrightarrow \mathbb{P}(T_Y)$.**

2. Main result & strategy towards the proofs

- The injection $0 \rightarrow T_X \rightarrow f^* T_Y$ induces the following diagram where Γ is the graph of the induced rational map $\mathbb{P}(f^* T_Y) \dashrightarrow \mathbb{P}(T_X)$.

$$\begin{array}{ccccc}
 & & \Gamma & & \\
 & \swarrow \beta & & \searrow \alpha & \\
 \mathbb{P}(T_Y) & \xleftarrow{\tilde{f}} & \mathbb{P}(f^* T_Y) & \dashrightarrow & \mathbb{P}(T_X) \\
 \downarrow \phi & & \downarrow \varphi & & \downarrow \tau \\
 Y & \xleftarrow{f} & X & \xlongequal{\quad} & X
 \end{array}$$

- Denote by ξ_X (resp. ξ_Y) the tautological line bundle of $\mathbb{P}(T_X)$ (resp. $\mathbb{P}(T_Y)$). Let $\tilde{\xi}_Y = \tilde{f}^* \xi_Y$ be the tautological line bundle of $\mathbb{P}(f^* T_Y)$.
- Let $\mathcal{K}$ (resp. $\mathcal{G}$) be a dominating family of minimal rational curves on X (resp. on Y). By assumption, the total dual VMRTs $D_X \subseteq \mathbb{P}(T_X)$ and $D_Y \subseteq \mathbb{P}(T_Y)$ are irreducible hypersurfaces.
- Let H be the ample generator of $\text{Pic}(X)$.

2. Main result & strategy towards the proofs

- As ξ_X is big, it follows from (Höring-Liu-Shao, 22), (Fu-Liu, 21) that $D_X \sim a\xi_X - b\tau^*H$ holds for $a > 0$ and $b > 0$.
- The total dual VMRT $D_X \subseteq \mathbb{P}(T_X)$ (a prime divisor by assumption) is extremal in the pseudo-effective cone $\text{PE}(\mathbb{P}(T_X))$.
- Since D_X is covered by minimal sections $\bar{\ell} \subseteq \mathbb{P}(T_X)$ of $\mathcal{K}$ such that $\xi_X \cdot \bar{\ell} = 0$, we have $D_X \cdot \bar{\ell} < 0$.

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- As ξ_X is big, it follows from (Höring-Liu-Shao, 22), (Fu-Liu, 21) that $D_X \sim a\xi_X - b\tau^*H$ holds for $a > 0$ and $b > 0$.
- The total dual VMRT $D_X \subseteq \mathbb{P}(T_X)$ (a prime divisor by assumption) is extremal in the pseudo-effective cone $\text{PE}(\mathbb{P}(T_X))$.
- Since D_X is covered by minimal sections $\bar{\ell} \subseteq \mathbb{P}(T_X)$ of $\mathcal{K}$ such that $\xi_X \cdot \bar{\ell} = 0$, we have $D_X \cdot \bar{\ell} < 0$.
- Let $D'_X := \beta_*(\alpha_*^{-1}(D_X))$ be the proper transform along the birational map $\mathbb{P}(f^*T_Y) \dashrightarrow \mathbb{P}(T_X)$. So we are left to show $D'_X = \tilde{f}^*D_Y$.

$$\begin{array}{ccccc}
 & & \Gamma & & \\
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 \end{array}$$

- From $0 \rightarrow T_X \rightarrow f^*T_Y$, we have the induced injection

$$0 \rightarrow H^0(X, \text{Sym}^a T_X \otimes \mathcal{O}_X(-bH)) \rightarrow H^0(X, \text{Sym}^a(f^*T_Y) \otimes \mathcal{O}_X(-bH)).$$

- As both α and β are birational with connected fibres, we have

$$0 \rightarrow H^0(\Gamma, a\alpha^*\xi_X - b\alpha^*\tau^*H) \rightarrow H^0(\Gamma, a\beta^*\widetilde{\xi_Y} - b\beta^*\varphi^*H).$$

2. Main result & strategy towards the proofs

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$$0 \rightarrow H^0(\Gamma, a\alpha^*\xi_X - b\alpha^*\tau^*H) \rightarrow H^0(\Gamma, a\beta^*\widetilde{\xi_Y} - b\beta^*\varphi^*H).$$

- Due to $D_X \sim a\xi_X - b\tau^*H$, there exists some $m \geq 0$ such that

$$a\widetilde{\xi_Y} + \varphi^*(-bH) \sim D'_X + m\varphi^*H.$$

So $D'_X \sim a\widetilde{\xi_Y} - (b+m)\varphi^*H$, with $a > 0, b > 0, m \geq 0$.

2. Main result & strategy towards the proofs

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 \downarrow \phi & & \downarrow \varphi & & \downarrow \tau \\
 Y & \xleftarrow{f} & X & \xlongequal{\quad\quad\quad} & X
 \end{array}$$

- Since D_X (as a prime divisor) is dominant over X , we see that D'_X is a prime divisor with $D'_X \sim a\widetilde{\xi_Y} - (b+m)\varphi^*H$.
- As the total dual VMRT D_Y is covered by minimal sections $\bar{c}$ such that $\xi_Y \cdot \bar{c} = 0$, $\tilde{f}^*D_Y$ is covered by curves $\bar{c}'$ such that $\widetilde{\xi_Y} \cdot \bar{c}' = 0$. So

$$\bar{c}' \cdot D'_X = \bar{c}' \cdot (a\widetilde{\xi_Y} - (b+m)\varphi^*H) < 0,$$

- Therefore, $\bar{c}' \subseteq D'_X$ and hence $\tilde{f}^*D_Y \subseteq D'_X$. By the irreducibility of D'_X , we obtain the equality.

3. A summary on the questions

Let X be a Fano manifold of Picard number 1.

- Describe the finite surjective morphisms $Y \rightarrow X$ from a smooth projective variety of Picard number 1. **Without assuming the positivity of T_X**
- Describe the finite surjective morphisms $X \rightarrow Z$ onto a smooth projective variety of Picard number 1. **Assuming the bigness of T_X**

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- Describe the finite surjective morphisms $X \rightarrow Z$ onto a smooth projective variety of Picard number 1. **Assuming the bigness of T_X**

Question

If the tangent bundle T_X is not pseudo-effective, can we obtain the inequality $c_n(\Omega_X(uH_X)) - u^n H_X^n > 0$ for certain (possibly away from finitely many points) globally generated $\Omega_X(uH_X)$?

If so, by an observation from (Amerik-Rovinsky-Van de Ven, 99); cf. (Shao-Z., 22), X satisfies Bounded-Degree-Conjecture. In (ARV, 99), (Beauville, 01), (SZ, 22), smooth hypersurfaces, special weighted hypersurfaces...have been verified.

Thank you for your attention!